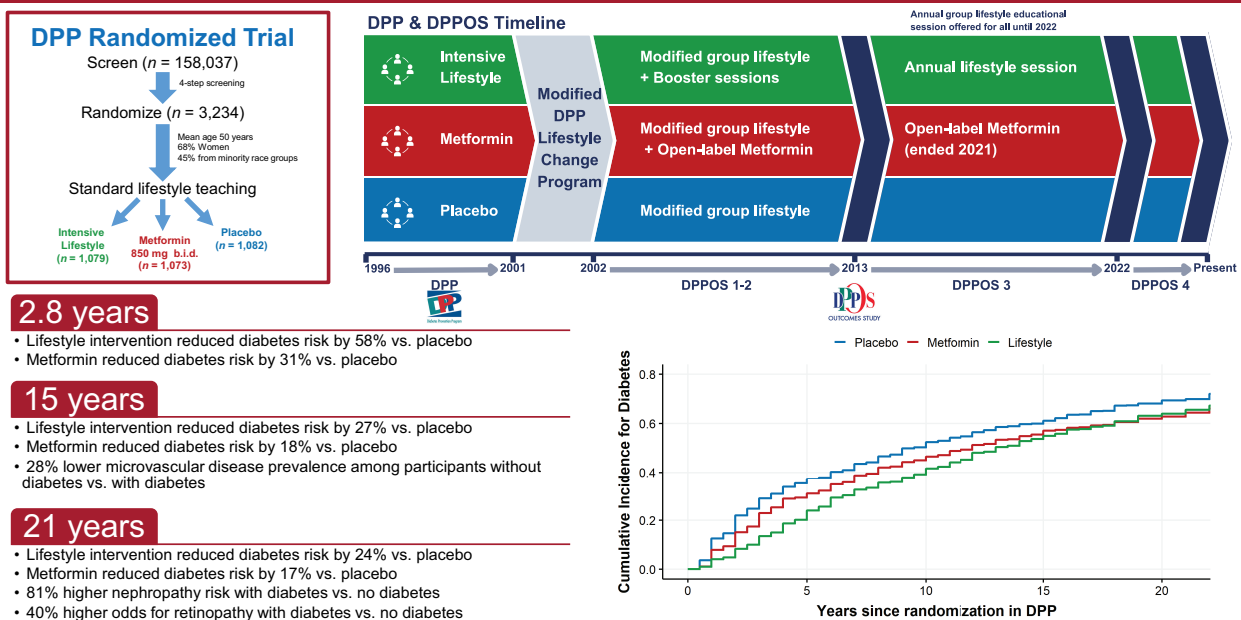


The Diabetes Prevention Program and Its Outcomes Study: NIDDK's Journey Into the Prevention of Type 2 Diabetes and Its Public Health Impact

Jill P. Crandall, Dana Dabelea, William C. Knowler, David M. Nathan, and Marinella Temprosa, on behalf of the DPP Research Group

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NIDDK's Journey in Type 2 Diabetes Prevention: Diabetes Prevention Effects With Lifestyle or Metformin Evident After 21 Years



ARTICLE HIGHLIGHTS

- Why did we undertake this study?**
Increasing prevalence of type 2 diabetes (T2D) and prevalence of its morbidities constitute major public health challenges.
- What is the specific question(s) we wanted to answer?**
Could an intensive lifestyle intervention or metformin reduce incidence of diabetes and related complications in a high-risk population?
- What did we find?**
After 2.8 years, diabetes risk was reduced 58% by intensive lifestyle intervention and 31% by metformin. Over 21 years of follow-up, median time to diabetes diagnosis decreased by 3.5 years (lifestyle intervention) and 2.5 years (metformin). Although there were no differences by randomized groups, among participants without diabetes there was lower prevalence of microvascular outcomes. Diabetes prevention effects were evident after 21 years.
- What are the implications of our findings?**
Prevention or delay of T2D with ILS or metformin is feasible and associated with improved health outcomes.



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on behalf of the DPP Research Group*

The current-day epidemic of type 2 diabetes, largely driven by increased adiposity and reduced physical activity in the setting of genetic susceptibility, is a major public health challenge. The National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) presciently proposed the Diabetes Prevention Program (DPP), a multi-center randomized clinical trial, designed by investigators in conjunction with NIDDK staff and initiated in 1996. The primary goal of DPP was to determine whether an intensive lifestyle intervention (ILS) or metformin in comparison with placebo would reduce the development of diabetes in a high-risk population with prediabetes. After mean 2.8 years, ILS reduced diabetes risk by 58% and metformin by 31%, leading to study termination ahead of schedule due to demonstrated efficacy of both interventions. In 2002, an extension of the DPP study, the Diabetes Prevention Program Outcomes Study (DPPOS), was initiated for examination of the longer-term course and consequences of diabetes prevention. Over 21 years of median total follow-up, in comparison with the placebo group, cumulative diabetes incidence was reduced by 24% and 17% in the original ILS and metformin groups, respectively, with median increases in diabetes-free survival of 3.5 and 2.5 years/person. During long-term follow-up, there were no significant effects of the original DPP interventions on microvascular or cardiovascular outcomes. However, compared with prevalence of microvascular outcomes among participants who progressed to diabetes, prevalence among those who did not progress was significantly lower. Longer-term follow-up of the cohort continues with examination of relationships between diabetes and prediabetes and an expanded array of diabetes- and aging-related morbidities.

The increasing prevalence of type 2 diabetes (T2D), largely driven by increased adiposity and reduced levels of physical activity in the setting of genetic risk factors, has become a major public health challenge during the past 40 years. In the U.S., diabetes increased progressively between 1990 and 2008, resulting in two million new cases diagnosed per year (1). The current (2021) estimated prevalence of T2D in the U.S. is >11% of adults (with ~30 million diagnosed cases) (2). The U.S. population at particularly high risk of developing diabetes, with prediabetes, totals another ~98 million. The number of cases of T2D worldwide now exceeds 800 million (3). The morbidity and mortality accompanying diabetes drive its human and economic costs; T2D is the major cause of blindness, kidney failure, and

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*A list of members of the DPP Research Group can be found in the supplementary material online.

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nontraumatic amputations in the U.S. T2D also increases the risk of nonspecific cardiovascular disease two- to fivefold.

Recognizing the major threat to public health, the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), National Institutes of Health (NIH), presciently initiated discussions in 1994 of a clinical trial to prevent diabetes. The successful conclusion of the Diabetes Control and Complications Trial in type 1 diabetes in 1993 (4) provided a template for NIDDK-sponsored large-scale, multicenter clinical trials. The Diabetes Prevention Program (DPP), a multicenter randomized clinical trial (RCT), was planned and developed by investigators in conjunction with NIDDK staff between 1994 and 1996 (5) (Fig. 1). The primary goal was to determine whether any of three active interventions (an intensive lifestyle intervention directed at weight loss and increased physical activity, metformin, or the thiazolidinedione troglitazone) would reduce the development of diabetes in comparison with the placebo control in adults at high risk of developing diabetes.

After the successful completion of the DPP (6), with a loyal cohort of participants, skilled research personnel at the 27 sites,

and a well-established Coordinating Center, central laboratories, and reading centers, a follow-up study was planned. The Diabetes Prevention Program Outcomes Study (DPPOS) was initiated in 2002 (7) and continues to the present for examination of the longer-term course and consequences of diabetes prevention (Fig. 1).

On the occasion of the 75th anniversary of the NIDDK, we review the major findings of the DPP and DPPOS. We focus on their major outcomes, established and projected consequences of diabetes prevention for long-term morbidity and mortality, the translation of our research findings into clinical practice and their public health implications, and the questions and challenges that remain. The DPP and DPPOS fulfill the NIDDK's mission "to conduct and support medical research and research training and to disseminate science-based information on diabetes . . . to improve people's health and quality of life."

WHY MIGHT T2D BE PREVENTABLE: PATHOPHYSIOLOGY AND POTENTIAL INTERVENTIONS

"Diabetes . . . is largely a penalty of obesity, and the greater the obesity, the

more likely is Nature to enforce it. The sooner this is realized by physicians and the laity, the sooner will the advancing frequency of diabetes be checked." (8). Joslin's 1923 simple summary of the pathogenesis of T2D remains true today. Although the possibility of preventing diabetes by addressing obesity had been acknowledged for many years, the first clinical trials to prevent T2D began in the 1960s with testing of drugs that stimulated insulin secretion or improved insulin sensitivity rather than targeting obesity directly. These trials lacked the size and rigor of modern RCTs and were of limited success and impact (9). Weight loss clinical trials followed.

How did understanding of the epidemiology and pathogenesis of T2D and these early RCTs lead to the DPP? A critical early development was the recognition by Himsworth in 1936 of two major types of diabetes (10), now known as type 1 (caused by autoimmune β -cell destruction) and type 2 (largely caused by obesity-mediated insulin resistance and impaired β -cell function). Preventing or reducing obesity was recognized as a key potential intervention to prevent T2D. Epidemiologic insights led to the recommendation 40 years ago that

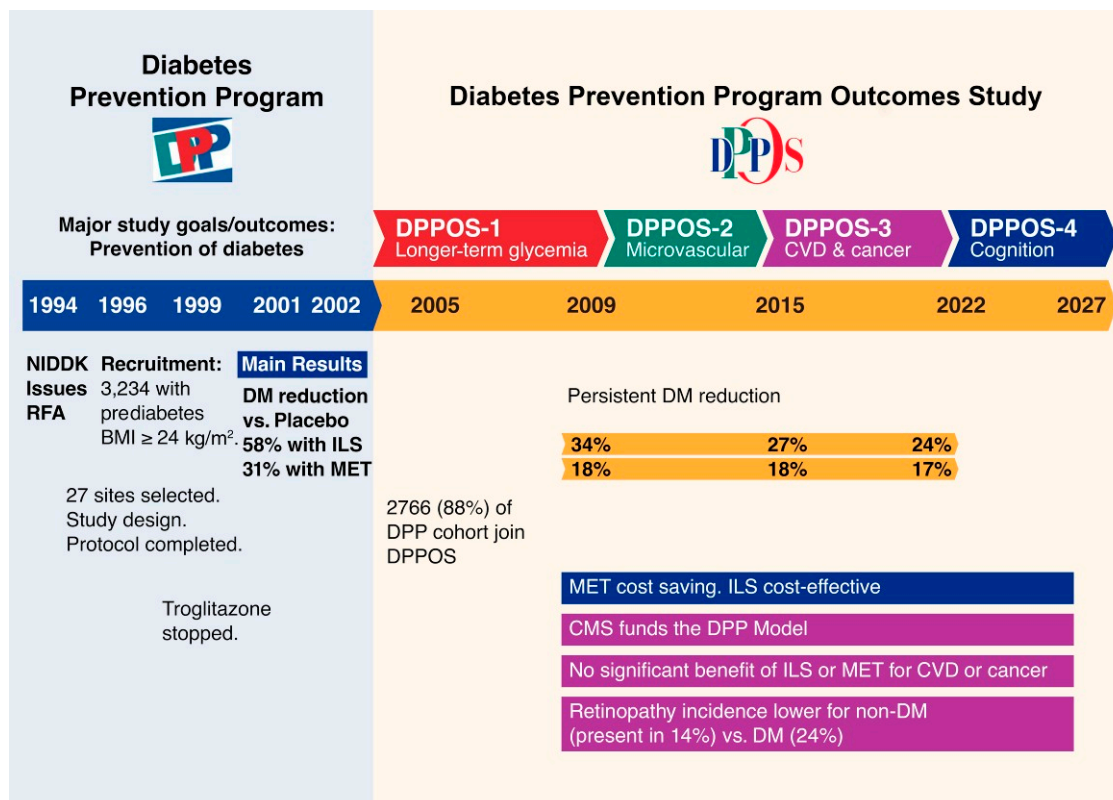


Figure 1—History of the DPP and DPPOS with milestones for aims, results, and impact. DM, diabetes mellitus; MET, metformin; RFA, request for applications.

“drugs, diet, or exercise” in individuals with impaired glucose tolerance be investigated for prevention of T2D (11). Treatment of individuals with impaired glucose tolerance was proposed to prevent subsequent β -cell failure and the development of T2D (12). The first two RCTs showing that weight loss through lifestyle changes could reduce the incidence of T2D, in China (1986–1992) (13) and Finland (1993–2000) (14), overlapped the DPP. Neither tested medications.

DPP: DESIGN, CONDUCT, AND MAJOR RESULTS

The DPP (1996–2001) was the first trial designed to evaluate the efficacy and safety of not only a lifestyle intervention aimed at weight loss and increased physical activity but also medications to prevent or delay the development of diabetes in individuals at high risk (5). Drug selection was controversial. Despite the putative importance of β -cell dysfunction, there were no available medications other than sulfonylureas that directly stimulated insulin secretion, and there were lingering concerns about their safety. Therefore, in the DPP two glucose-lowering medications were selected, metformin and the thiazolidinedione troglitazone, for testing in parallel with a lifestyle intervention. Metformin improved hepatic insulin sensitivity (15) and troglitazone improved insulin sensitivity in muscle (16). Although neither medication had a known direct effect on β -cells, through improvement of insulin sensitivity, demand for insulin secretion would be reduced and β -cell function theoretically preserved.

More than 158,000 individuals were screened for identification of eligible adults, age ≥ 25 years, at high risk of developing diabetes. Eligibility criteria included overweight or obesity (BMI ≥ 24 or ≥ 22 kg/m² for Asian Americans); elevated fasting plasma glucose (FPG) 95–125 mg/dL (5.3–6.9 mmol/L), or ≤ 125 mg/dL (6.9 mmol/L) in the American Indian centers; and impaired glucose tolerance defined as 2-h glucose 140–199 mg/dL (7.8–11 mmol/L) after a 75-g oral glucose tolerance test (OGTT). Other eligibility criteria have previously been described in detail (5).

Ethics Approval and Consent to Participate

The study was conducted in accordance with the guidelines established in the Declaration of Helsinki and followed all required procedures for research involving human participants. Prior to initiation of the study protocol, each participant provided written informed consent and each study center obtained approval from its respective institutional review board. The trials are registered at ClinicalTrials.gov (DPP clinical trial reg. no. NCT00004992, DPPOS NCT00038727, and DPPOS AD/ADRD NCT05704309).

Eligible participants ($n = 3,819$) were randomly assigned (1996–1999) to metformin (850 mg b.i.d.), troglitazone (400 mg q.d.), double placebo, or intensive lifestyle intervention (ILS) (5,6,17). All treatment groups were double masked except for ILS. The troglitazone arm was discontinued in 1998 due to emerging concerns with liver toxicity (17), leaving a three-arm study group of 3,234 participants (Fig. 1). Troglitazone was subsequently withdrawn from the U.S. market. By design, the study population was diverse with regard to ethnic and racial distribution, age, and sex. More than 45% of the cohort self-identified as of a minority ethnic or racial group, and 68% were women. At randomization, average age was 51 years and 20% were ≥ 60 years old. Average BMI was 34 kg/m², 28% had hypertension, and 7% were current smokers (6,18) (Table 1).

The primary outcome, diabetes development, was assessed semiannually on the basis of FPG or annually with OGTT according to ADA criteria of FPG ≥ 126 mg/dL or 2-h glucose ≥ 200 mg/dL, with confirmation. In metformin and placebo group participants diagnosed with diabetes, study metformin or placebo was continued until hyperglycemia progressed to FPG ≥ 140 mg/dL (confirmed), at which time participants were unmasked and referred to their own physician for diabetes treatment. Intensive lifestyle modification was reinforced among ILS group participants who developed diabetes and referred for treatment. All participants who developed diabetes continued to be followed.

After a mean of 2.8 years (range 1.8–4.6) of study, the independent data monitoring board recommended in May 2001 that the trial stop, 1 year ahead of schedule, based on the demonstrated

efficacy of both interventions. Compared with placebo, the lifestyle intervention had reduced body weight by ~ 4 kg and diabetes development by 58%. Adherence to metformin was $\sim 70\%$, and it reduced diabetes by 31% ($P < 0.001$ for ILS and metformin groups compared with placebo) (6) (Fig. 2). The effects of lifestyle intervention and metformin did not differ by age, sex, or race and ethnicity (Fig. 3). Metformin was effective only in participants with BMI ≥ 35 kg/m² and in those with fasting glucose 110–125 mg/dL at baseline.

Of note, even though troglitazone was stopped early owing to safety concerns, subsequent analyses of the 387 participants who took it for a mean of 0.9 years revealed a significant 75% reduction in diabetes incidence (17). The benefit dissipated rapidly after discontinuation of troglitazone, and whether it would have continued to have a beneficial effect is unknown.

MECHANISMS OF DIABETES PREVENTION

The lifestyle intervention led to reduced weight, increased self-reported physical activity, and improved estimated insulin sensitivity and secretion (6,19). The reduction in diabetes incidence in the ILS group was largely attributable to weight loss, with a 16% reduction in diabetes development per kilogram lost (20). Metformin was significantly less effective than the lifestyle intervention but also led to improved estimated insulin sensitivity and insulin response (19), albeit to a lesser extent than with the lifestyle intervention. Sixty-four percent of metformin's effect on diabetes risk relative to placebo was attributable to the weight loss induced by the drug (21). Although neither weight loss nor metformin has known direct effects on β -cell function, both interventions improved estimated insulin response relative to insulin sensitivity (19,22), consistent with the hypothesis that improving insulin sensitivity would “off-load” the strain on the β -cell. The amount of weight loss was directly associated with decreased estimated insulin secretory demand and increased compensatory secretory response, which in turn mediated the preventive effects of metformin and the lifestyle interventions (22).

Table 1—Characteristics for DPP baseline, start of DPPOS, and end of DPPOS-3

	DPP baseline (1996–1999): all participants	DPPOS start (2002–2003)*			DPPOS (2019–2020)+		
		Placebo	Metformin	ILS	Placebo	Metformin	ILS
N randomized	3,234	1,082	1,073	1,079	1,082	1,073	1,079
N attended visit	3,234	888	891	866	679	674	658
Median follow-up (years)	—	5	5	5	21	21	21
Age (years)	50.6 ± 10.7	54.8 ± 10.0	55.5 ± 10.1	55.3 ± 11.0	70.5 ± 9.1	71.2 ± 8.9	71.0 ± 9.7
Female sex	2,191 (68)	642 (72)	617 (69)	619 (71)	492 (72)	467 (69)	463 (70)
Race and ethnicity							
Non-Hispanic White	1,768 (55)	501	515	490	363	352	331
African American	645 (20)	192	191	176	132	154	129
Hispanic	508 (16)	145	141	138	104	103	108
American Indian	171 (5)	54	46	53	48	42	45
Asian American	142 (4)	53	31	40	32	23	45
Fasting glucose (mg/dL)	107 ± 8	113 ± 25	107 ± 16	109 ± 29	128 ± 38	123 ± 39	128 ± 38
HbA _{1c} (%)	5.90 ± 0.50	6.01 ± 0.79	5.94 ± 0.63	5.89 ± 0.64	6.50 ± 1.30	6.30 ± 1.20	6.50 ± 1.30
BMI (kg/m ²)	34.0 ± 6.7	33.9 ± 7.1	33.0 ± 6.8	32.9 ± 7.1	32.6 ± 6.8	31.8 ± 6.6	31.8 ± 6.6
Systolic BP	124 ± 15	123 ± 15	124 ± 15	122 ± 14	125 ± 14	124 ± 14	124 ± 14
Diastolic BP	78 ± 9	75 ± 9	75 ± 9	75 ± 9	71 ± 10	70 ± 10	70 ± 10
Diabetes‡	0	324 (36)	272 (31)	171 (20)	473 (70)	425 (63)	428 (65)

Data are *n* (%) or mean ± SD unless otherwise indicated. BP, blood pressure. *Characteristics of participants who attended their first DPPOS annual or midyear visit. +Characteristics of participants who attended DPPOS 17A visits, which represented the last complete annual visits. ‡Diabetes data reflect number and proportion with diabetes diagnosis among those who attended the visit.

By contrast, troglitazone, the major known effect of which is to increase insulin sensitivity, resulted in increased body weight. However, during its relatively brief use in DPP, it reduced the development of diabetes significantly more than metformin and placebo and nominally more than the lifestyle intervention (17).

These results suggest that weight loss may not be a required component of T2D prevention, although redistribution of body fat may be important.

A genetic contribution to T2D has long been recognized, but the genetic basis is very complex, with an increasing number of genetic variants associated

with risk. One of the strongest identified genetic risk factors is variation at the *TCF7L2* gene, the effect of which was confirmed in DPP participants (23). Importantly, the effects of the lifestyle and metformin interventions were greater in those with the high-risk *TCF7L2* variants, suggesting that people with greater

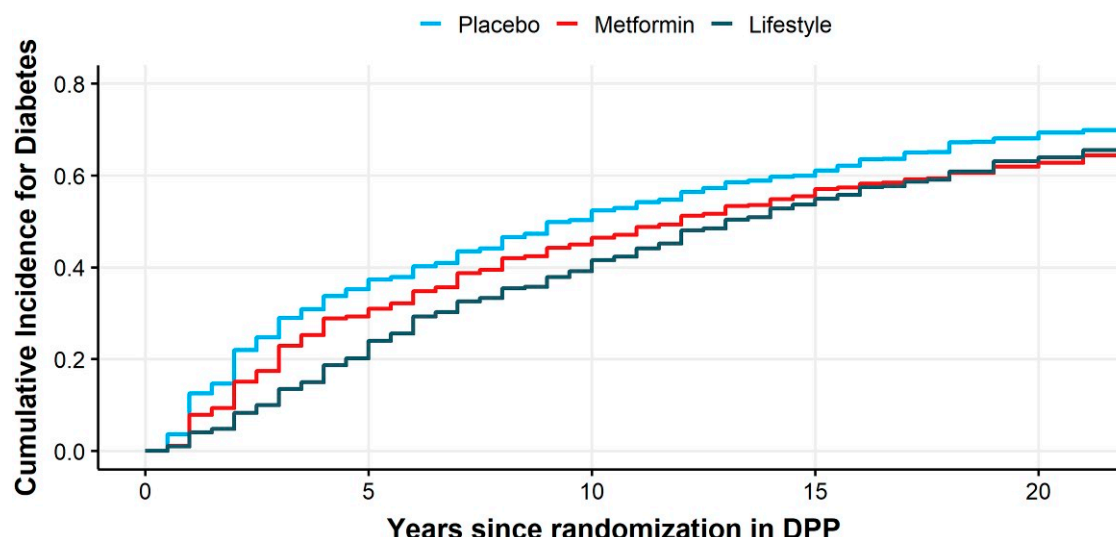


Figure 2—Cumulative incidence of diabetes during DPP and DPPOS (31). The DPP (1996–2002) was conducted for an average of 2.8 years (range 1.8–4.6), after which DPP participants were invited to join the DPPOS (2003–present).

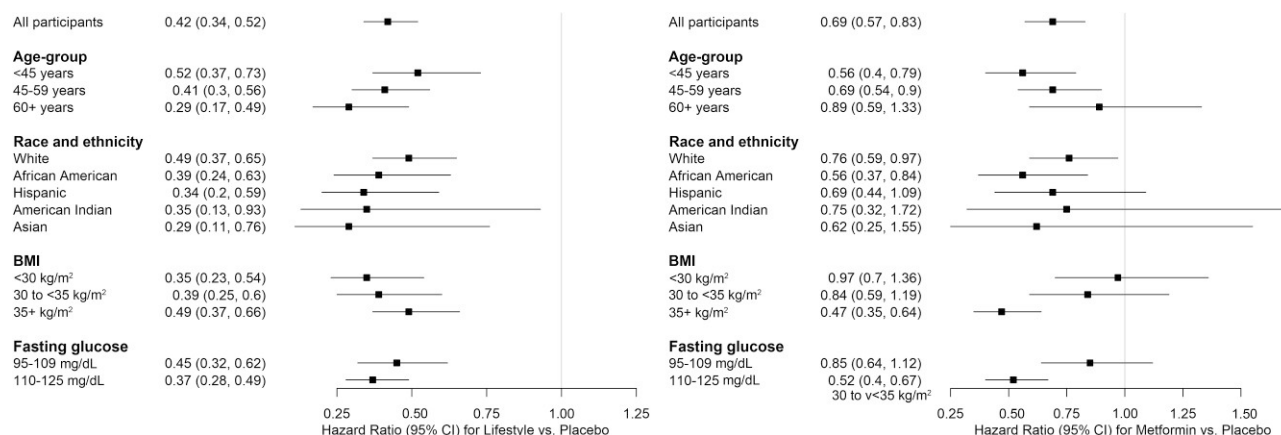


Figure 3—Effects of lifestyle and metformin on diabetes development by subgroups categorized according to baseline measurements (6). The solid squares and lines represent point and 95% CI of the estimated HRs for the effects of ILS vs. placebo (left) and metformin vs. placebo (right). There were significant interactions ($P < 0.05$) between metformin and BMI and between metformin and fasting glucose. None of the other subgroup-by-treatment interactions were statistically significant.

genetic susceptibility might derive more, rather than less, benefit from prevention efforts. Subsequent genetics analyses in DPP confirmed the effects of many other previously identified variants on risk for diabetes and that the interventions were generally effective regardless of genotype (24).

The primary DPP results are consistent with those of other RCTs of lifestyle intervention (13,14,25), of metformin interventions (25), and of troglitazone or other medications in the thiazolidinedione class, as previously reviewed (9). In recent, usually short-term clinical trials, other medications, including glucagon-like peptide 1 (GLP-1) receptor agonists, have reduced the incidence of T2D (26,27).

DPPOS

The value of diabetes prevention as a public health strategy lies not merely in the prevention/delay of diabetes development but also in the potential to reduce diabetes-associated long-term morbidity and mortality. DPP-based prevention programs have been endorsed by public health entities, but there is debate over the cost-benefit of screening and treating individuals with “prediabetes,” since some will never progress to diabetes and resources might be better directed at those with established disease (28–30).

The long-term extension of the DPP during DPPOS (2002–present) (7) was undertaken to determine whether the original DPP interventions resulted in sustained prevention of diabetes and whether diabetes prevention reduced the development of related chronic

complications, which are generally a product of diabetes duration and degree of hyperglycemia. Outcomes to date have included the following: further development of diabetes and evolution of dysglycemia; occurrence of microvascular complications (retinopathy, nephropathy, and neuropathy); aging and diabetes-related comorbidities such as cardiovascular disease (CVD), cancer, cognitive impairment and dementia, physical function, and quality of life and economic outcomes; and mortality.

TRANSITION FROM DPP TO DPPOS

During a 1-year bridge period following DPP completion, 88% ($n = 2,766$) of active participants (including those who had been diagnosed with diabetes) enrolled in DPPOS. The intensive lifestyle intervention was offered to all study participants but now in a group format (7). Placebo was discontinued and metformin treatment was continued, but unmasked, in the original randomized treatment group until 2021. Quarterly lifestyle (“HELP”) sessions provided ongoing support for all participants, and those originally assigned to the ILS group were also offered supplemental lifestyle (“Boost”) campaigns twice a year. In metformin group participants diagnosed with diabetes, study metformin was continued until hyperglycemia progressed to $HbA_{1c} \geq 7\%$, at which time they were referred to their own physician for diabetes treatment.

Semiannual FPG and annual HbA_{1c} measurements were continued in all participants, and annual OGTTs were performed in those who had not been

diagnosed with diabetes. Assessment of diabetic nephropathy (urine albumin-to-creatinine ratio in a spot urine sample and serum creatinine for calculation of estimated glomerular filtration rate [eGFR]) and CVD risk factors was continued, as was formal adjudication of CVD events and cause of mortality. New outcomes, including assessment of diabetic retinopathy with fundus photography, were added. Given the aging of the cohort during long-term follow-up, study outcomes were broadened to include those affecting older adults, such as physical and cognitive function and frailty, for better understanding of the impact of diabetes and diabetes prevention.

LONG-TERM EFFECTS ON DIABETES AND EVOLUTION OF DYSGLYCEMIA

During 21 years of median follow-up (as of February 2020), in comparison with placebo, diabetes incidence was reduced by lifestyle intervention by 24% (95% CI 15, 32) and by metformin by 17% (7,26) (Fig. 2), with increases in median diabetes-free survival of 3.5 and 2.5 years/person, respectively (31). These reductions translated into 21-year cumulative incidences of 70% (67,73), 64% (61,68), and 66% (62,69) in the original placebo, metformin, and ILS groups, respectively. The findings of long-term effects of the interventions on cumulative incidence of diabetes resulted largely from early effects on annual incidence rates during the active DPP intervention period, with little or no additional effect during the long-term follow-up in DPPOS. Similar results for long-term effects

of a lifestyle intervention in Chinese adults with impaired glucose tolerance were reported after 30 years of follow-up in the Da Qing study (32).

In DPPOS, intervention effects were greater with lifestyle intervention in subgroups at highest risk (with higher baseline fasting glucose and HbA_{1c} and advanced age) with little or no effect in those at lowest risk of diabetes (31). With metformin, greater intervention effects were seen in younger participants and among women with a history of gestational diabetes mellitus. Metformin was ineffective in those ≥ 60 years old at baseline. These findings indicate that effective prevention efforts can result in sustainable reductions in diabetes burden for several decades. In addition, the demonstrated heterogeneity of diabetes prevention effects can be leveraged in future precision prevention efforts.

Beyond prevention or delay of the development of diabetes, the long-term effects of restoration of normal glucose regulation (NGR) were explored in DPPOS (33). Diabetes risk during 10 years of follow-up was 56% lower for participants who had returned to NGR at least once during the active intervention period than for those who consistently had prediabetes (hazard ratio [HR] 0.44, [95% CI 0.37, 0.55]), independent of the original DPP intervention. Thus, early, measures

aimed at restoring NGR in high-risk people may be important approaches to reducing future diabetes risk.

MICROVASCULAR COMPLICATIONS

At 15 years of follow-up (as of January 2014), there were no significant effects of the original DPP interventions on the aggregate microvascular outcome, composed of nephropathy, neuropathy, and retinopathy (34) (Table 2). However, in comparison with participants who progressed to diabetes, among those who did not progress there was a 28% lower prevalence of microvascular complications ($P < 0.0001$).

Retinopathy

Retinopathy was assessed with seven-field stereoscopic fundus photography obtained during DPPOS and diagnosed according to Early Treatment Diabetic Retinopathy Study (ETDRS) grade ≥ 20 in either eye (35) or treatment with laser or intravitreal injections. At 21 years of follow-up, 24% of the participants who had developed diabetes had retinopathy, while only 14% of those without diabetes had developed retinopathy (odds ratio [OR] 1.47 [95% CI 1.22, 1.79]) (36). However, original DPP intervention was not associated with retinopathy (37) or with age-related macular degeneration (38). Presence of retinopathy was also

associated with duration of diabetes at the time of fundus examination (1.30 per SD [1.18, 1.44]) and higher mean HbA_{1c}, fasting and 2-h plasma glucose, systolic and diastolic blood pressure, and weight during follow-up, as well as with current smoking (36). In multivariate analyses, mean HbA_{1c} during follow-up continued to be significantly associated with retinopathy (1.65 per SD 0.7% [1.48, 1.83]) across the entire glycemic range from prediabetes to diabetes.

Peripheral Neuropathy

Peripheral neuropathy was measured annually with a sensory exam and the Michigan Neuropathy Screening Instrument (MNSI) questionnaire (39). At 21 years of median follow-up, as with retinopathy, there was a higher likelihood of peripheral neuropathy associated with diabetes status (OR vs. no diabetes 1.40 [95% CI 1.07, 1.84]), longer duration of diabetes (OR 1.04 per year [1.02, 1.05]), and higher average HbA_{1c} (OR 1.85 per 1% increase [1.54, 2.21]) (40). The presence of peripheral neuropathy did not differ across the original DPP interventions.

Nephropathy

Nephropathy was measured annually and defined according to albuminuria ≥ 30 mg/g creatinine in a spot urine

Table 2—Incidence and prevalence of diabetes-related complications by randomized group and by diabetes status

	Placebo	Metformin	ILS	Metformin vs. placebo	ILS vs. placebo	Diabetes vs. no diabetes
Incident events (reference no.)						
Nephropathy (41)	145 (10.0)	152 (10.2)	141 (10.8)	1.08 (0.86, 1.35)	1.02 (0.81, 1.29)	1.81 (1.43, 2.30)
MACE (45)	98 (5.28)	101 (5.51)	111 (6.10)	1.03 (0.78, 1.37)	1.14 (0.87, 1.50)	1.06 (0.84, 1.35)
MACEplus (45)	157 (8.73)	157 (8.86)	174 (9.93)	1.00 (0.80, 1.25)	1.12 (0.90, 1.39)	1.08 (0.89, 1.30)
All cancers (53)	191 (10.8)	173 (9.8)	182 (10.5)	0.90 (0.73, 1.10)	0.96 (0.79, 1.18)	0.98 (0.82, 1.17)
Obesity-related cancer (53)	99 (5.4)	88 (4.8)	91 (5.1)	0.89 (0.66, 1.18)	0.93 (0.70, 1.23)	1.08 (0.84, 1.39)
Mortality (46)	143 (6.59)	152 (7.13)	158 (7.37)	0.99 (0.79, 1.25)	1.02 (0.81, 1.28)	1.06 (0.88, 1.28)
DPPOS prevalent events (reference no.)						
Retinopathy at Y16 (36,37)	8.8	9.1	10.0	1.03 (0.68, 1.57)	1.15 (0.76, 1.75)	1.47 (1.22, 1.79)
DSPN at Y17 (40)	21.9	21.5	21.5	0.92 (0.69, 1.22)	0.95 (0.71, 1.26)	1.40 (1.07, 1.84)
Frailty at Y10 (52)	5.4	5.3	3.6	0.99 (0.69, 1.42)	0.62 (0.42, 0.93)	1.46 (1.03, 2.07)
Microvascular disease at Y11 (34)	12.4	13.0	11.3	0.91 (0.78, 1.07)	1.05 (0.91, 1.23)	0.72 (0.63, 0.83)

Data for incident events are no. of events (event rate/1,000 patient-years) or HR (95% CI). Data for DPPOS prevalent events are prevalence (%) or OR (95% CI), except for microvascular disease at year 11 (Y11), for which data are prevalence (%) or relative risk (95% CI). For incident event analyses Cox models were used with adjustment for age, sex, race and ethnicity, and randomized group including data from all participants ($n = 3,234$) except for nephropathy, restricted to those with eGFR ≥ 45 mL/min/1.73 m² and albumin-to-creatinine ratio < 30 at baseline ($n = 2,574$). MACE included fatal and nonfatal myocardial infarction and stroke. MACEplus was defined as MACE plus hospitalization for unstable angina or heart failure, revascularization for coronary or peripheral vascular disease, coronary artery disease diagnosed by angiography, and/or "silent" myocardial infarction diagnosed by electrocardiogram. Nephropathy was defined according to eGFR < 45 mL/min/1.73 m² or albumin-to-creatinine ratio ≥ 30 and retinopathy according to ETDRS ≥ 20 (34). Distal symmetrical polyneuropathy (DSPN) was defined according to abnormal sensory exam or symptoms (MNSI [38]). Frailty was determined according to Fried frailty phenotype (50). Microvascular disease was defined based on aggregate outcome composed of nephropathy, retinopathy, and neuropathy at DPPOS year 11 (34).

collection or estimated GFR <45 mL/min/ 1.73 m², with the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation (41), both on two consecutive tests, or renal failure (end-stage renal disease, dialysis, or transplantation). Across 21 years of median follow-up, the risk for developing nephropathy was higher among participants with diabetes in comparison with those without (HR 1.81 [95% CI 1.43, 2.30]) but did not differ by intervention group (41).

Regression to NGR during DPP was associated with lower odds of the aggregate microvascular outcome and of retinopathy and nephropathy individually, as well as during DPPOS, in models adjusted for age, sex, race and ethnicity, baseline HbA_{1c}, and original DPP intervention (OR 0.78 [95% CI 0.65, 0.78]) (42). This effect was due to lower glycemic exposure over time. These results support the importance of diabetes prevention as an approach to reducing the burden of costly microvascular complications.

CVD RISK FACTORS, CVD OUTCOMES, AND ALL-CAUSE MORTALITY

Analyses were conducted to explore the effects of original DPP interventions on CVD risk factors, including trajectories of mean LDL cholesterol, triglyceride, systolic blood pressure, BMI, HDL cholesterol, and HbA_{1c} levels and percent receiving statin and blood pressure-lowering medications. Similar to observations during DPP (43) and earlier phases of DPPOS (44), during 21 years of median follow-up, CVD risk factors were generally more favorable in the ILS and metformin groups than in the placebo group, except for LDL cholesterol and urine albumin levels and glomerular filtration rate, which did not differ over time by original intervention group (45). Overall, blood pressure-lowering medication and statin use were common (56%–74%), with no significant differences across treatment groups.

CVD outcomes were adjudicated by the Outcomes Committee, masked to treatment assignment and following standard methodology using medical records, death certificates, autopsy reports, and research records. The first occurrence of a major adverse cardiovascular event (MACE), defined as nonfatal myocardial infarction (excluding silent myocardial infarction),

nonfatal stroke, or fatal CVD, was used as the main outcome. An extended CVD event outcome (extended-MACE) was comprised of the first occurrence of MACE or hospitalization for congestive heart failure or unstable angina, coronary or peripheral revascularization, coronary heart disease diagnosed by angiography, or silent myocardial infarction (45).

During the 21 years of median follow-up, 310 participants experienced a first MACE. The incidence of MACE did not differ by original DPP intervention: metformin versus placebo HR 1.03 (95% CI 0.78, 1.37) and ILS versus placebo 1.14 (0.87, 1.50) (45). Among the MACE components, there were nominally fewer nonfatal strokes in the metformin than in the placebo group, with no significant difference in rates (HR 0.57 [0.31, 1.06]) and no significant differences between treatment groups in the other two MACE outcomes, nonfatal myocardial infarctions and cardiovascular death. Similarly, there were no significant associations between interventions and the incidence of the extended-MACE outcome. Prespecified subgroup analyses on intervention effects on MACE showed no significant heterogeneity by age, sex, race and ethnicity, or diabetes status for either metformin or ILS.

These data suggest that, despite significant long-term reduction in diabetes development and improvement in CVD risk factors, metformin and lifestyle interventions may not have additional effects on MACE in the setting of prediabetes or recent-onset, well-controlled diabetes, especially in a population with high use of blood pressure-lowering and statin medications and lower CVD rates than the national average. Thus, metformin and lifestyle intervention may not provide additional protection against CVD once glycemia, lipids, and blood pressure are well controlled.

All-cause mortality was ascertained through regular surveillance at annual visits and two National Death Index searches, and cause of death was adjudicated by a committee blinded to treatment assignment (46). Over a median 21 years of follow-up, 453 participants died. Cancer was the leading cause of death ($n = 170$), followed by CVD ($n = 131$). Compared with placebo, neither metformin nor lifestyle intervention influenced all-cause mortality (HR 0.99 [95% CI 0.79, 1.25] for metformin, 1.02 [0.81, 1.28] for ILS),

cancer mortality (1.04 [0.72, 1.52] for metformin, 1.07 [0.74, 1.55] for ILS), or CVD mortality (1.08 [0.70, 1.66] for metformin, 1.18 [0.77, 1.81] for ILS). Analyses adjusted for diabetes and duration, BMI, cumulative glycemic exposure, and cardiovascular risk factors yielded similar results.

Cognition

Diabetes is increasingly recognized as an important risk factor for cognitive decline and dementia (47). There have been conflicting reports about the effects of metformin on cognition (48,49). As the longest randomized metformin study, DPP/DPPOS was uniquely positioned for exploration of the effects of metformin as well as those of lifestyle modification on cognition. Cognition was assessed with tests of memory (Spanish-English Verbal Learning Test) and executive function (Digit Symbol Substitution Test and animal and letter fluency tests) in DPPOS year 8 (~12 years after randomization), when the average age of participants was 63 ± 10 years (SD), and 2 years later. No treatment group differences in cognitive scores (individual tests or a composite z score) were observed, and there was no association with diabetes, although higher HbA_{1c} at the time of cognitive testing was related to worse cognitive performance in age-adjusted models (50). Cumulative metformin exposure, including randomized treatment and out-of-study treatment for diabetes, was unrelated to cognitive scores. A limitation of these analyses was that most participants were younger when tested than the age when cognitive decline typically occurs. The ongoing DPPOS phase 4 (DPPOS-4) is devoted to further study of cognitive decline and incident dementia as the cohort enters the age span when they are most vulnerable. In DPPOS-4 neuroimaging and measurement of biomarkers are performed over time to identify the causes of cognitive impairment.

Frailty

Frailty is a common geriatric syndrome of physiologic decline, characterized by increased vulnerability to poor health outcomes, including falls, hospitalization, disability, and death. Diabetes and obesity are accepted as important risk factors for frailty, and we hypothesized that ILS and

diabetes prevention could modify this risk. Frailty assessment with the Fried frailty phenotype criteria (51) was conducted in DPPOS years 8 and 10. In year 10, the prevalence of frailty was lowest in the ILS group (3.6%) and similar in the metformin (5.3%) and placebo (5.4%) groups, with OR 0.62 (95% CI 0.42, 0.93) for ILS compared with placebo (52). Overall, participants who had developed diabetes were almost 50% more likely to be frail (OR 1.46 [1.03, 2.07]) than those who remained without diabetes. These results suggested that in individuals at high risk of diabetes, an ILS intervention in middle age could reduce frailty in later life.

Cancer

The increasing recognition of obesity and T2D as risk factors for cancer and interest in the potential anticancer effects of metformin led to the ascertainment and adjudication of incident cancers, in collaboration with the National Cancer Institute, after a median follow-up of 21 years (53). In an intention-to-treat analysis, no significant difference was seen in incident cancers (total or obesity related) between treatment groups, although cancer incidence rates were nominally lowest in the metformin group (HR 0.90 [95% CI 0.73, 1.10], metformin versus placebo). Diabetes, mean HbA_{1c}, and metformin exposure were not related to incident cancers in time-dependent multivariate models.

HEALTH-RELATED QUALITY OF LIFE

Prevention of diabetes would be expected to have a salutary effect on health-related quality of life, assessed in DPP/DPPOS with the 36-Item Short Form Health Survey (SF-36) (54). At completion of DPP, those randomized to ILS, but not metformin, had small but significant improvements in the physical component summary (PCS) score ($P < 0.0001$) and health utility index (SF-6D) ($P = 0.04$), compared with the placebo group (55). In the ILS group, initial improvements were seen in several domains, including general health, physical function, bodily pain, and vitality, which were mediated by successful weight loss and increased physical activity (55). Health-related quality of life scores tended to decline during DPPOS in all treatment groups, regardless of incident diabetes (56).

COST-EFFECTIVENESS OF LIFESTYLE INTERVENTION AND METFORMIN

The DPPOS has also conducted one of the few long-term cost-effectiveness analyses of diabetes prevention. Over 10 years of follow-up, lifestyle intervention was cost-effective, and metformin was cost saving in comparison with placebo (57). From a payer's perspective, investment in lifestyle and metformin interventions for diabetes prevention in high-risk adults provides good value.

TRANSLATION AND IMPACT AT INDIVIDUAL AND POPULATION LEVELS

In the wake of the consistent demonstration that diabetes could be delayed or prevented in the DPP (6), the Da Qing study in China (13), and the Finnish Diabetes Prevention Study (14), among others (9,58), numerous studies addressed a variety of methods that might make the lifestyle interventions more widely available and could promulgate the implementation of lifestyle programs (59). In the majority of these short-term studies investigators examined the achievement of the DPP weight loss and physical activity goals rather than the prevention or delay of diabetes per se. They did demonstrate that DPP lifestyle goals could be achieved, using fewer expert staff involved than in DPP and with mobile/electronic applications and in settings such as YMCAs (60). Following the DPP findings, the European Association for the Study of Diabetes in 2004 (61), International Diabetes Federation in 2007 (62), and American Diabetes Association in 2008 (63) recommended lifestyle programs and metformin for prevention of diabetes in high-risk populations. After the demonstration of cost-efficacy, as required by the Centers for Medicare and Medicaid Services (CMS), CMS began to fund the implementation of the "Diabetes Prevention Program Model" in 2018 (64). Governmental funding for diabetes prevention programs was initiated in China, the U.K., and other countries around the world over time (65). Metformin received regulatory approval for an indication "to help prevent T2D if you're at high risk" in the U.K. in 2017 (66) and in other countries, although not in the U.S.

PUBLIC HEALTH IMPACT

The reduction in diabetes development during the DPP and the longer-term effects demonstrated during DPPOS could be projected for populations with similar characteristics. Unfortunately, in the absence of accurate national registry data, assessment of the impact over time of the DPP/DPPOS findings on diabetes incidence can only be speculative. Of note, after a steady increase in diabetes incidence and prevalence during the period preceding the DPP (between the 1980s and 2008), there have been signs of the incidence of diabetes in U.S. adults decreasing, between 2008 and 2021 (1,2). Whether the decreasing incidence is real has been questioned (67).

LIMITATIONS

Although DPP and DPPOS have demonstrated clinically important short- and long-term effects of the interventions and other benefits of diabetes prevention, including economic benefits, the limitations of the studies should be noted. First, the changes in therapy during the transition to DPPOS make the interpretation of the long-term results somewhat challenging. The provision of the lifestyle intervention to the metformin and placebo groups, and a less intensive lifestyle program provided to the original DPP ILS group, may have reduced the relative effects of lifestyle over time in the intention-to-treat analyses. Second, as with all clinical trials the eligibility criteria of the study limit generalizability of the results. Finally, although the population size was large enough to provide good power to demonstrate the benefits of the interventions regarding diabetes development, the size of the relatively low-risk cohort may have been too small for demonstration of benefits for some of the diabetes-associated outcomes.

FUTURE OF DIABETES PREVENTION

As important as the DPP/DPPOS results are in preventing or delaying the onset of T2D in U.S. adults, they are not good enough. Over time, most of the participants, who were selected because they were at high risk, ultimately developed T2D. What new research directions might lead to improved results? For individuals

identified in the prediabetes stage, like those who enrolled in DPP, more effective and lasting weight loss interventions should help. These might come from improved behavioral intervention methods; more effective, long-term weight loss medications; or metabolic surgery. In a recent 3-year clinical trial of individuals with prediabetes and obesity, the GIP/GLP-1 receptor agonist tirzepatide dramatically reduced the incidence of diabetes (26). The GLP-1 receptor agonist semaglutide has also been shown to reduce the development of diabetes, in a 3-year trial in participants with overweight or obesity with preexisting CVD but without diabetes at baseline (68). Other medications could prevent T2D through direct effects on insulin resistance, such as the thiazolidinedione drugs, or on β -cell preservation, although long-term effectiveness and safety of such drugs would need to be demonstrated.

Perhaps more effective, although more difficult to implement and evaluate, would be interventions starting earlier in life in individuals not meeting current criteria for prediabetes. Genetic and epidemiologic studies have shown that susceptibility to obesity and T2D begins at conception and is strongly influenced by the intrauterine and early childhood environments, such that obesity and T2D are increasingly developing during childhood (69). This is especially true among racial and ethnic groups at highest risk for T2D (70–72). We must better understand and manage the early-life determinants of T2D.

Individuals live in societies that strongly influence their behaviors and environments. Many aspects of modern American society promote sedentary behavior and obesity, making it difficult for individuals to optimize their behaviors for diabetes prevention. Societal changes will be necessary for better management of the obesity epidemic and diabetes prevention.

SUMMARY AND CONCLUSIONS

In the DPP, among adults at high risk for T2D, both ILS and metformin group interventions prevented or delayed the development of diabetes during 3 years of active placebo-controlled treatments. The DPPOS long-term follow-up study, during which group-implemented lifestyle intervention was offered to all participants and metformin was continued until 2021 in the original metformin

group, showed that although annual diabetes incidence rates declined overall, the treatment group separation in diabetes cumulative incidence rates persisted over 21 years. This resulted in increased diabetes-free survival in both intervention groups in comparison with the original placebo group. While lifestyle intervention prevented or delayed diabetes in all age-groups studied, metformin appeared to be ineffective in those age ≥ 60 years at baseline.

Incidence of diabetes-related complications, including retinopathy, peripheral neuropathy, nephropathy, and CVD, was lower in participants who did not develop diabetes during long-term follow-up. However, comparisons between the randomized treatment groups have yielded largely null results in terms of diabetes complications and associated comorbidities. This “paradox” can be explained by a number of factors, an important one being the minimal separation in glycemia and other vascular risk factors between treatment groups following diabetes diagnosis (73). This is likely a consequence of the ethical study design, as risk factors (e.g., blood pressure, lipids, HbA_{1c}) were frequently assessed in all participants and shared with care providers with treatment recommendations, resulting in a high level of risk mitigation. In addition, a modified version of the original ILS group intervention was offered to all participants and nonstudy metformin use became increasingly common as participants developed diabetes, both leading effectively to crossover between the original treatment groups. Finally, the diminished intensity of the original interventions over time and the long period between the interventions and the outcomes of interest, factors that take many years to develop, may also minimize detectable treatment group differences. The durable and clinically relevant delay in the onset of diabetes and the uniform reduction of diabetes-specific complications when diabetes development was delayed or NGR was restored remain the key legacies of DPP/DPPOS.

Current evidence for effective and safe means of preventing T2D long-term supports lifestyle intervention, with an emphasis on weight loss and moderate physical activity, at any adult age, and supports metformin in younger adults. The relative benefits of lifestyle intervention and drug interventions may change with recent and future research on newer

medicines (26) that control hyperglycemia or body weight and with improvements in long-term maintenance of behavioral weight loss. With 30 years of support from the NIDDK over its 75-year history and fulfilling NIDDK’s mission to improve public health through prevention and treatment, the DPP and DPPOS have shown the potential of both lifestyle and metformin to reduce the burden of T2D. Further progress in preventing T2D will require 1) improvements in screening and individual-directed preventive interventions; 2) addressing determinants of T2D earlier in life (including prenatally), given the increase in youth-onset T2D; and 3) societal measures to modify the diabetogenic environment.

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Data and Resource Availability. In accordance with the NIH Public Access Policy, we continue to provide all manuscripts to PubMed Central including this manuscript. For DPP/DPPOS the protocols and lifestyle and medication intervention manuals have been provided to the public through its public website (<https://dppos.bsc.gwu.edu>). The DPPOS abides by the NIDDK data sharing policy and implementation guidance as required by the NIH/NIDDK (<https://repository.niddk.nih.gov/study/40>).

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